THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **World of Numbers** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** $\frac{7}{9}$	Each is $\frac{d}{9}$.
2	**(b)** $\frac{4}{9}$	Add the numerators over 99.
3	**(a)** 3	Cancel every prime factor other than 2 and 5.
4	**(b)** 4	$\frac{1}{x} = 2 - \sqrt{3}$, so the sum is 2×2.
5	**(c)** $\frac{14}{11}$	Write as $1 + 0.\overline{ab}$.
6	**(a)** 1	Reason about decimal expansions.
7	**(b)** 5	Find the order of 10 modulo q.
8	**(a)** 2	Multiply the fraction by 11 and reduce.
9	**(a)** Both A and R are true and R is the correct explanation of A.	$0.375 = \dfrac{375}{1000} = \dfrac{3}{8}$; R is the general reason.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both statements are true, but R is a property of irrational numbers and does not prove that $\sqrt{2}$ is irrational.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$0.\overline{17} = \frac{17}{99}$, $0.\overline{71} = \frac{71}{99}$ (1); Sum $= \frac{8}{9}$ (1).	1+1
12	$420 = 2^2 \times 3 \times 5 \times 7$ (1); Cancel the factors other than 2 and 5: $n = 21$ (1).	1+1
13	$\frac{1}{x} = \frac{6-\sqrt{35}}{6^2-35} = 6 - \sqrt{35}$ (1); $x + \frac{1}{x} = 12$ (1).	1+1
14	$0.\overline{29} = \frac{29}{99}$, $0.\overline{92} = \frac{92}{99}$ (1); Sum $= \frac{11}{9}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$x = 0.4\overline{7}$: $100x - 10x = 43$ (1); $x = \frac{43}{90}$ (1); Long division of p by q gives $0.4\overline{7}$ (1).	1+1+1
16	$\frac{1}{x} = 2 - \sqrt{3}$ since $2^2 - 3 = 1$, so $x + \frac{1}{x} = 4$ (1); $x^2 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)^2 - 2$ (1); $= 16 - 2 = 14$ (1).	1+1+1
17	Suppose $3 + \sqrt{2} = r$ is rational; Then $\sqrt{2} = r - 3$ (1); $r - 3$ is rational, so $\sqrt{2}$ would be rational (1); This contradicts the given fact (1).	1+1+1
	OR: $x = 5.3\overline{2}$: $100x - 10x = 479$ (1); $x = \frac{479}{90}$ (1); Long division of p by q gives $5.3\overline{2}$ (1).	1+1+1
18	$\frac{1}{x} = 3 - \sqrt{8}$ since $3^2 - 8 = 1$, so $x + \frac{1}{x} = 6$ (1); $x^2 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)^2 - 2$ (1); $= 36 - 2 = 34$ (1).	1+1+1
	OR: Suppose $5 + \sqrt{3} = r$ is rational; Then $\sqrt{3} = r - 5$ (1); $r - 5$ is rational, so $\sqrt{3}$ would be rational (1); This contradicts the given fact (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\dfrac{7}{16} = 0.4375$ (1). (ii) $\dfrac{11}{25} = 0.44$ (1). (iii) $12 = 2^2 \times 3$ (1). (iv) $\dfrac{7}{16}$ and $\dfrac{11}{25}$ terminate (1) because $16 = 2^4$ and $25 = 5^2$ have only the prime factors 2 and 5; $\dfrac{5}{12} = 0.41\overline{6}$ does not terminate because 12 has the factor 3 (1).	1+1+1+2
20	Divide 5 by 13 until a remainder repeats (1); $\frac{5}{13} = 0.\overline{384615}$ (1); Block 384615 (1); Block length $= 6$ (1); Repeating, so the number is rational (1).	1+1+1+1+1
	OR: First $= 2(11 + 3) = 28$ (1); Second $= 11 - 3 = 8$ (1); Both are rational (1); In the first, the cross terms $\pm 2\sqrt{33}$ cancel (2).	1+1+1+2

Mistakes to avoid:

- Testing the denominator of $\dfrac{p}{q}$ for factors 2 and 5 before reducing the fraction to lowest terms.
- Believing the product (or sum) of two irrational numbers is always irrational: $\sqrt{2} \times \sqrt{8} = 4$.
- Treating $\dfrac{22}{7}$ or 3.14 as the exact value of π; both are rational approximations, π itself is irrational.
- Saying a decimal is irrational because it is long or non-terminating; a repeating decimal such as $0.\overline{142857}$ is rational.

- Using $0.\overline{ab} = \frac{ab}{100}$ instead of $\frac{ab}{99}$ when converting a recurring decimal.
- Getting the sign wrong for debt $\times$ debt: the product of two negatives is a fortune (positive).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **World of Numbers** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 49	Cancel every prime factor other than 2 and 5.
2	**(a)** 6	$\frac{1}{x} = 3 - \sqrt{8}$, so the sum is 2×3.
3	**(a)** $\frac{15}{11}$	Write as $1 + 0.\overline{ab}$.
4	**(b)** irrational	Reason about decimal expansions.
5	**(b)** 3	Find the order of 10 modulo q.
6	**(a)** 3	Multiply the fraction by 11 and reduce.
7	**(a)** $\frac{10}{9}$	Each is $\frac{d}{9}$.
8	**(b)** 1	Add the numerators over 99.
9	**(c)** A is true but R is false.	$(\sqrt{3})^2 - (\sqrt{2})^2 = 3 - 2 = 1$, so A is true; R is false because this product is rational.
10	**(d)** A is false but R is true.	A is false because 7 has a prime factor other than 2 and 5; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$90 = 2 \times 3^2 \times 5$ (1); Cancel the factors other than 2 and 5: $n = 9$ (1).	1+1
12	$\frac{1}{x} = \frac{7-\sqrt{48}}{7^2-48} = 7 - \sqrt{48}$ (1); $x + \frac{1}{x} = 14$ (1).	1+1
13	$0.\overline{36} = \frac{36}{99}$, $0.\overline{63} = \frac{63}{99}$ (1); Sum $= 1$ (1).	1+1
14	$1430 = 2 \times 5 \times 11 \times 13$ (1); Cancel the factors other than 2 and 5: $n = 143$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$x = 6.7\overline{9}$: $100x - 10x = 612$ (1); $x = \frac{34}{5}$ (1); Long division of p by q gives 6.8 (1).	1+1+1
16	$\frac{1}{x} = 4 - \sqrt{15}$ since $4^2 - 15 = 1$, so $x + \frac{1}{x} = 8$ (1); $x^2 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)^2 - 2$ (1); $= 64 - 2 = 62$ (1).	1+1+1
17	Suppose $7 + \sqrt{5} = r$ is rational; Then $\sqrt{5} = r - 7$ (1); $r - 7$ is rational, so $\sqrt{5}$ would be rational (1); This contradicts the given fact (1).	1+1+1
	OR: $x = 0.9\overline{4}$: $100x - 10x = 85$ (1); $x = \frac{17}{18}$ (1); Long division of p by q gives $0.9\overline{4}$ (1).	1+1+1
18	$\frac{1}{x} = 5 - \sqrt{24}$ since $5^2 - 24 = 1$, so $x + \frac{1}{x} = 10$ (1); $x^2 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)^2 - 2$ (1); $= 100 - 2 = 98$ (1).	1+1+1
	OR: Suppose $4 + \sqrt{6} = r$ is rational; Then $\sqrt{6} = r - 4$ (1); $r - 4$ is rational, so $\sqrt{6}$ would be rational (1); This contradicts the given fact (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\sqrt{2}$ m (1). (ii) $\sqrt{8} = 2\sqrt{2}$ m (1). (iii) $\sqrt{2} \times 2\sqrt{2} = 4$ (1). (iv) $5 + \sqrt{2}$ is irrational because it is the sum of a rational and an irrational number (1); $(\sqrt{2}+1)(\sqrt{2}-1) = 2 - 1 = 1$, so the product of two irrational numbers can be rational (1).	1+1+1+2
20	Divide 10 by 37 until a remainder repeats (1); $\frac{10}{37} = 0.\overline{270}$ (1); Block 270 (1); Block length $= 3$ (1); Repeating, so the number is rational (1).	1+1+1+1+1
	OR: First $= 2(10 + 6) = 32$ (1); Second $= 10 - 6 = 4$ (1); Both are rational (1); In the first, the cross terms $\pm 2\sqrt{60}$ cancel (2).	1+1+1+2

Mistakes to avoid:

- Testing the denominator of $\frac{p}{q}$ for factors 2 and 5 before reducing the fraction to lowest terms.
- Believing the product (or sum) of two irrational numbers is always irrational: $\sqrt{2} \times \sqrt{8} = 4$.
- Treating $\frac{22}{7}$ or 3.14 as the exact value of π; both are rational approximations, π itself is irrational.
- Saying a decimal is irrational because it is long or non-terminating; a repeating decimal such as $0.\overline{142857}$ is rational.

- Using $0.\overline{ab} = \frac{ab}{100}$ instead of $\frac{ab}{99}$ when converting a recurring decimal.
- Getting the sign wrong for debt $\times$ debt: the product of two negatives is a fortune (positive).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **World of Numbers** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** $\frac{16}{11}$	Write as $1 + 0.\overline{ab}$.
2	**(c)** infinitely many rational numbers	Reason about decimal expansions.
3	**(a)** 6	Find the order of 10 modulo q.
4	**(a)** 5	Multiply the fraction by 11 and reduce.
5	**(c)** $\frac{5}{9}$	Each is $\frac{d}{9}$.
6	**(a)** $\frac{11}{9}$	Add the numerators over 99.
7	**(a)** 11	Cancel every prime factor other than 2 and 5.
8	**(d)** 10	$\frac{1}{x} = 5 - \sqrt{24}$, so the sum is 2×5.
9	**(b)** Both A and R are true but R is not the correct explanation of A.	$\frac{3}{8} = 0.375$ terminates, so A is true; R is true, but not every rational number terminates, so R does not explain A.
10	**(a)** Both A and R are true and R is the correct explanation of A.	3 is rational and $\sqrt{5}$ is irrational, so R applies directly.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\frac{1}{x} = \frac{8-\sqrt{63}}{8^2-63} = 8 - \sqrt{63}$ (1); $x + \frac{1}{x} = 16$ (1).	1+1
12	$0.\overline{58} = \frac{58}{99}$, $0.\overline{85} = \frac{85}{99}$ (1); Sum $= \frac{13}{9}$ (1).	1+1
13	$504 = 2^3 \times 3^2 \times 7$ (1); Cancel the factors other than 2 and 5: $n = 63$ (1).	1+1
14	$\frac{1}{x} = \frac{9-\sqrt{80}}{9^2-80} = 9 - \sqrt{80}$ (1); $x + \frac{1}{x} = 18$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$x = 7.1\overline{8}$: $100x - 10x = 647$ (1); $x = \frac{647}{90}$ (1); Long division of p by q gives $7.1\overline{8}$ (1).	1+1+1
16	$\frac{1}{x} = 6 - \sqrt{35}$ since $6^2 - 35 = 1$, so $x + \frac{1}{x} = 12$ (1); $x^2 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)^2 - 2$ (1); $= 144 - 2 = 142$ (1).	1+1+1
17	Suppose $2 + \sqrt{7} = r$ is rational; Then $\sqrt{7} = r - 2$ (1); $r - 2$ is rational, so $\sqrt{7}$ would be rational (1); This contradicts the given fact (1).	1+1+1
	OR: $x = 8.2\overline{3}$: $100x - 10x = 741$ (1); $x = \frac{247}{30}$ (1); Long division of p by q gives $8.2\overline{3}$ (1).	1+1+1
18	$\frac{1}{x} = 7 - \sqrt{48}$ since $7^2 - 48 = 1$, so $x + \frac{1}{x} = 14$ (1); $x^2 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)^2 - 2$ (1); $= 196 - 2 = 194$ (1).	1+1+1
	OR: Suppose $9 + \sqrt{10} = r$ is rational; Then $\sqrt{10} = r - 9$ (1); $r - 9$ is rational, so $\sqrt{10}$ would be rational (1); This contradicts the given fact (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $4 \times (-6) = -24$, a debt of Rs 24 (1). (ii) $-7 - 0 = -7$ (1). (iii) $(-3) \times (-8) = 24$, a fortune (1). (iv) $20 + (-18) = 2$ (1), which is a fortune of 2 (1).	1+1+1+2
20	Divide 12 by 101 until a remainder repeats (1); $\frac{12}{101} = 0.\overline{1188}$ (1); Block 1188 (1); Block length $= 4$ (1); Repeating, so the number is rational (1).	1+1+1+1+1
	OR: First $= 2(19 + 7) = 52$ (1); Second $= 19 - 7 = 12$ (1); Both are rational (1); In the first, the cross terms $\pm 2\sqrt{133}$ cancel (2).	1+1+1+2

Mistakes to avoid:

- Testing the denominator of $\frac{p}{q}$ for factors 2 and 5 before reducing the fraction to lowest terms.
- Believing the product (or sum) of two irrational numbers is always irrational: $\sqrt{2} \times \sqrt{8} = 4$.
- Treating $\frac{22}{7}$ or 3.14 as the exact value of π; both are rational approximations, π itself is irrational.
- Saying a decimal is irrational because it is long or non-terminating; a repeating decimal such as $0.\overline{142857}$ is rational.

- Using $0.\overline{ab} = \frac{ab}{100}$ instead of $\frac{ab}{99}$ when converting a recurring decimal.
- Getting the sign wrong for debt $\times$ debt: the product of two negatives is a fortune (positive).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND